Introduction to Polymer Materials

Spring, 2022

SUN Dazhi

Syllabus

- **Reference Books**

  Polymer Science and Technology Joel R. Fried
- **Lectures**
- TA: 赵静
- **Related labs in** <<材料学综合实验>>
- QQ group

Syllabus

- **Grading**

Homework: 20%

Term paper: 20%

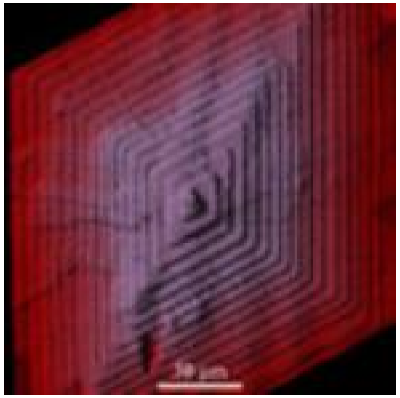
Midterm exam: 20%

Final exam: 40%

Additional score： 3%

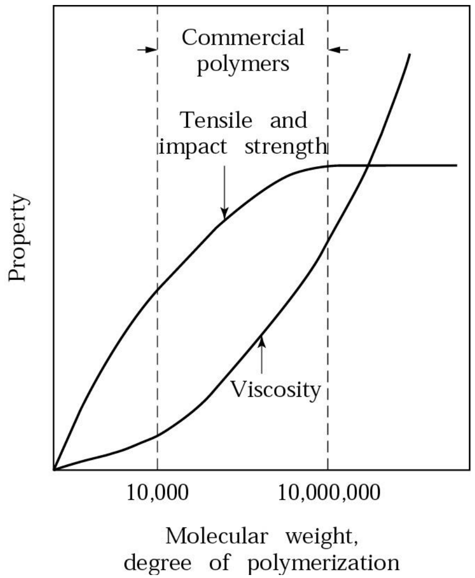
- maximum 2 talks before each class

Materials Science of Polymers for Engineers

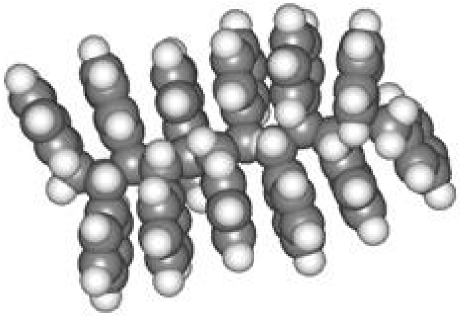


Physics

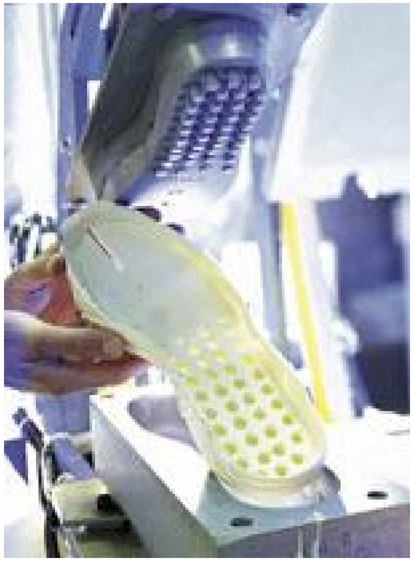
Applications



Properties



Chemistry



Processing

Major Functions of Polymers

Adhesives

superglue

epoxies *polyethylene*

polyesters

Barriers

Polyethylene landfill
Garbage bags
Sarah wrap

Structural components

PPMA or PC
transparent sheets
Molded ABS or HIPS

Insulation

Polyurethane foam
Styrofoam
Polyethylene wire coatings
Bakelite (phenol-formaldehyde)

Goals

Learn:

- Basic polymer nomenclature and structure
- Basic types of polymers and how they are made
- Properties of polymers
- Solution properties
- Polymer processing
- Aging & degradation of polymers
- Applications of polymers
- Understand where polymers should be used and what their limitations are.
- Frontier in polymers
- *How to communicate*
- *How to think skeptically*

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- Start paper early
- Study groups
- Prepare exams
- Don’t cheat, plagiarize, or otherwise participate in un-ethical behavior
- Learn with Internet: Wikipedia, google(baidu)…
- Ask questions
- Think skeptically

Thinking skeptically

- Don’t trust anyone
- If it doesn’t make sense, *ask questions*.
- Beware of trusting experts, textbooks, and online information.
- Acquaint yourself with logic and logical fallacies

Term Paper

- *Review of literature topic*
- *Petition* to present topic relating to research
- *JACS* style bibliography, >= 5 pages
- electronic copy
- Literature review on any topics related to polymers
- **Make a video to introduce your topics（mid-term?）**
- One draft due during semester

Lecture 1: Introduction

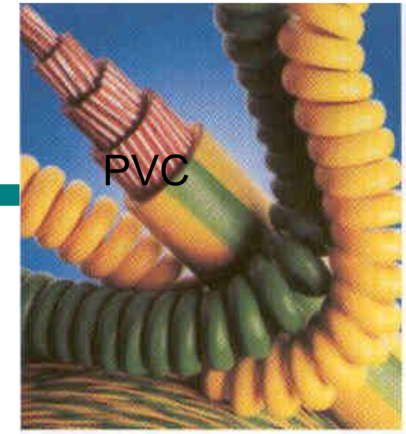
SUN Dazhi

Polymers are everywhere

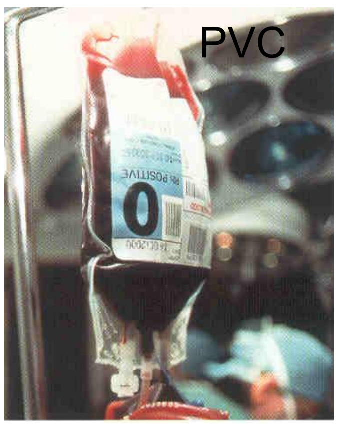
Food Packaging

Transportation

Polyisoprene

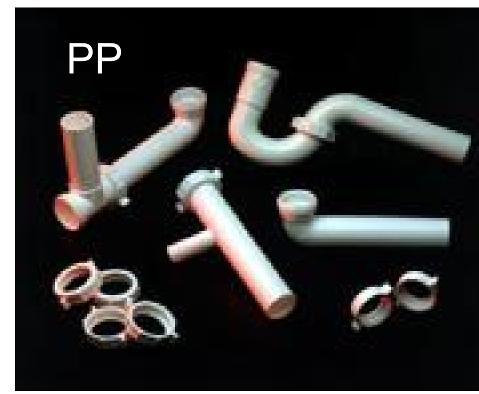


Electronics

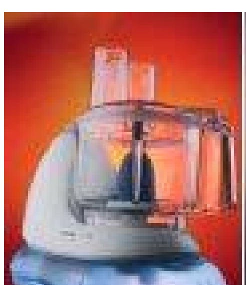


Medical Supplies

Clothing



Construction



Manufactured Goods

PC

We use a lot of polymers

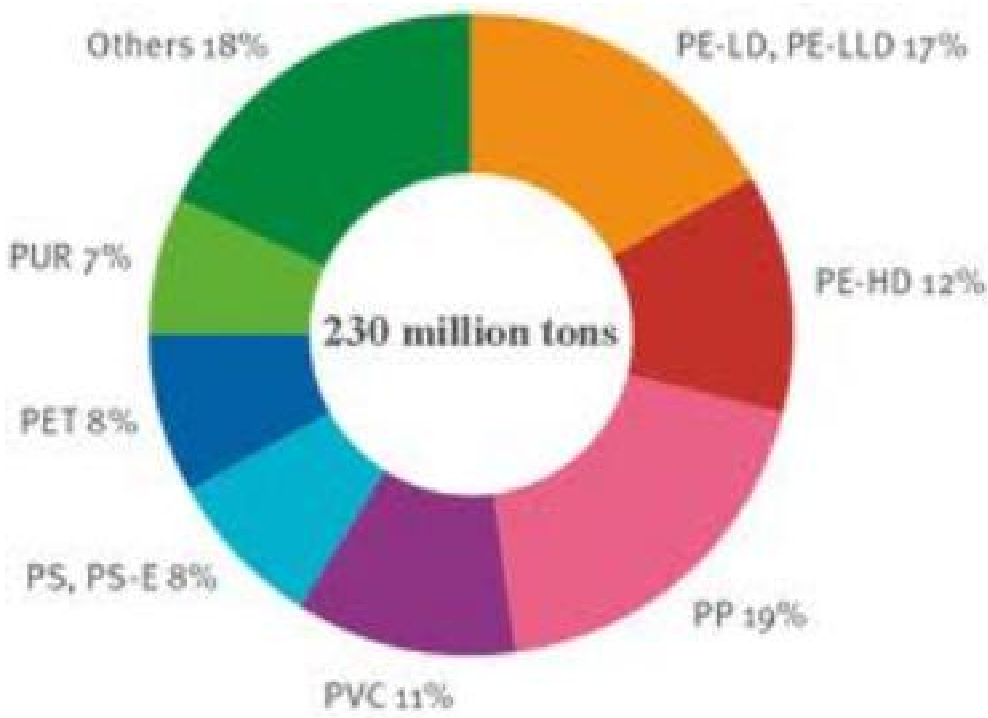
Polymer Resins; Millions of pounds '95

COMMODITY RESINS (CR)		ENGINEERING THERMOPLASTICS	
LDPE	7,500	PC	760
LLDPE	6,000	ACETALS	330
HDPE	11,600	PAI	
EVA	1,200	ARYLATES	
PP	10,600	PEEK	
PVC	11,800	PEI	
ACRYLICS	550	PI	
PS	6,000	PPO	
ABS	1,500	PPS	
PET	4,000	SULPHONE	
POLYESTER	1,600	LCP	
NYLON	1,000		
Total	63,350		

> 50% of CR crystallizable POs.
~70% of total CR crystallizable.

10^{12} bags/year!!

We use a lot of polymers

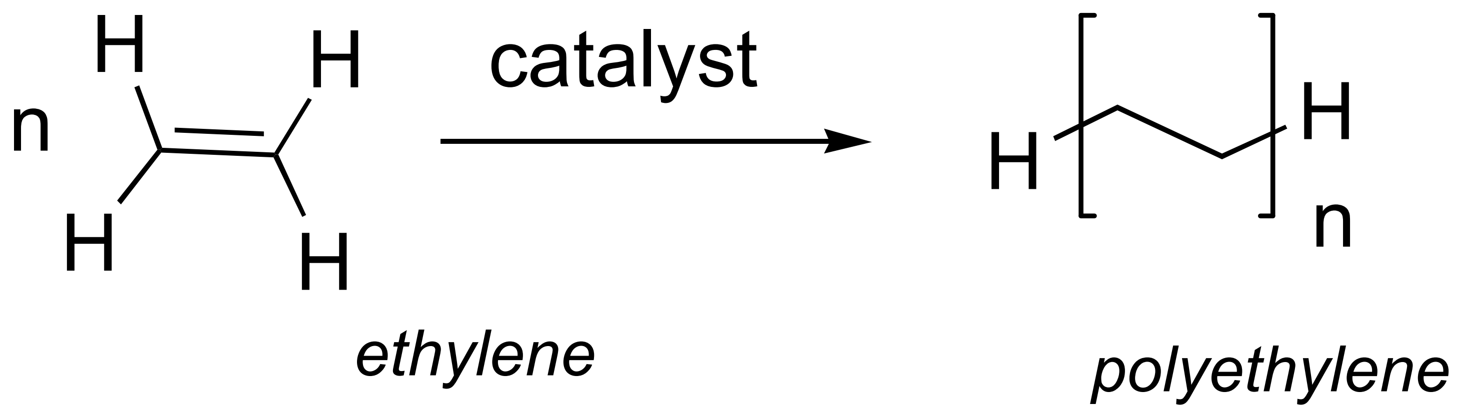


World wide consumption of plastics in 2009
~9% increase annually

What are polymers?

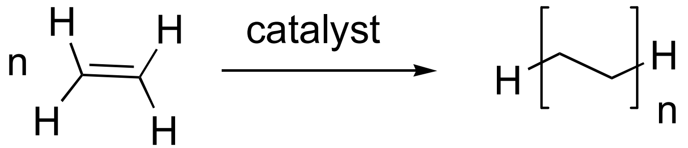
Poly = many & meros = parts (Greek)

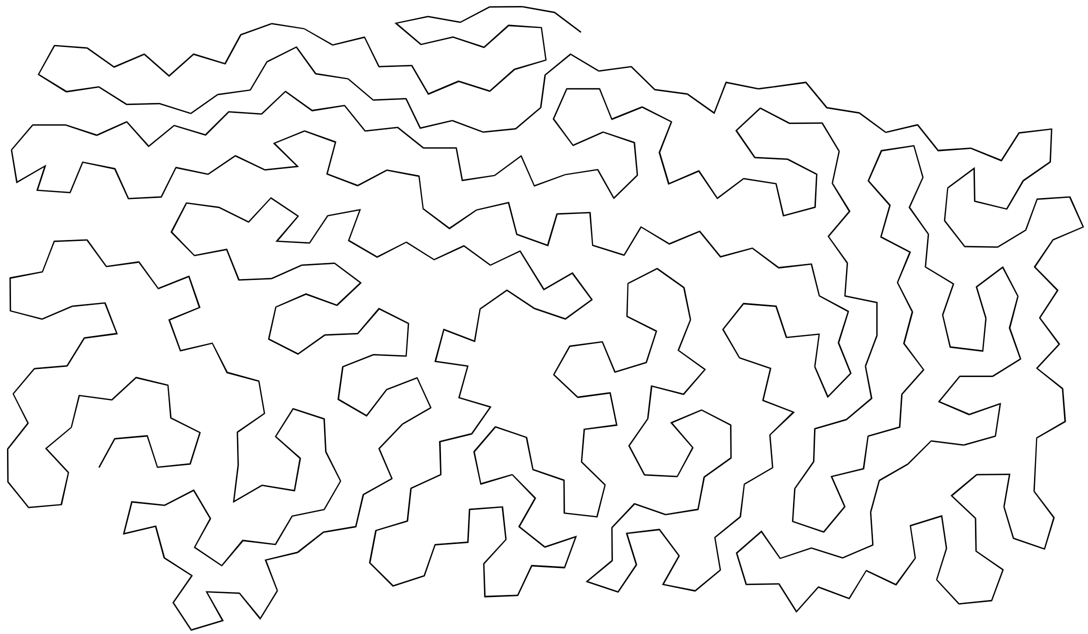
Macromolecules = large molecules



A nomenclature exists to describe polymers

What are polymers?



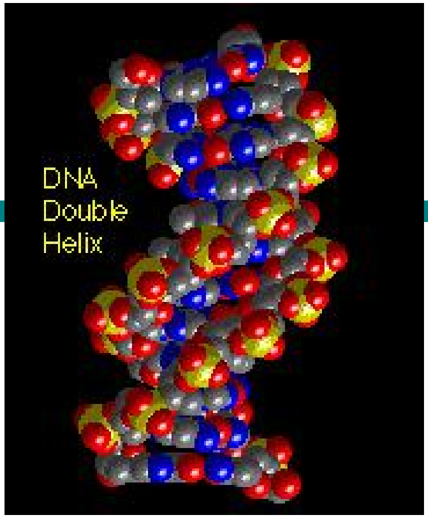


Chemical Formula: $C_{500}H_{1002}$
Molecular Weight: 7015.31
Elemental Analysis: C, 85.60; H, 14.40

Contour length: 38.5 nm or 0.0385 microns or 0.0000385 mm

10^6 Dalton polyethylene (35.7K monomers) = 5.5 microns or 0.0055 mm in length

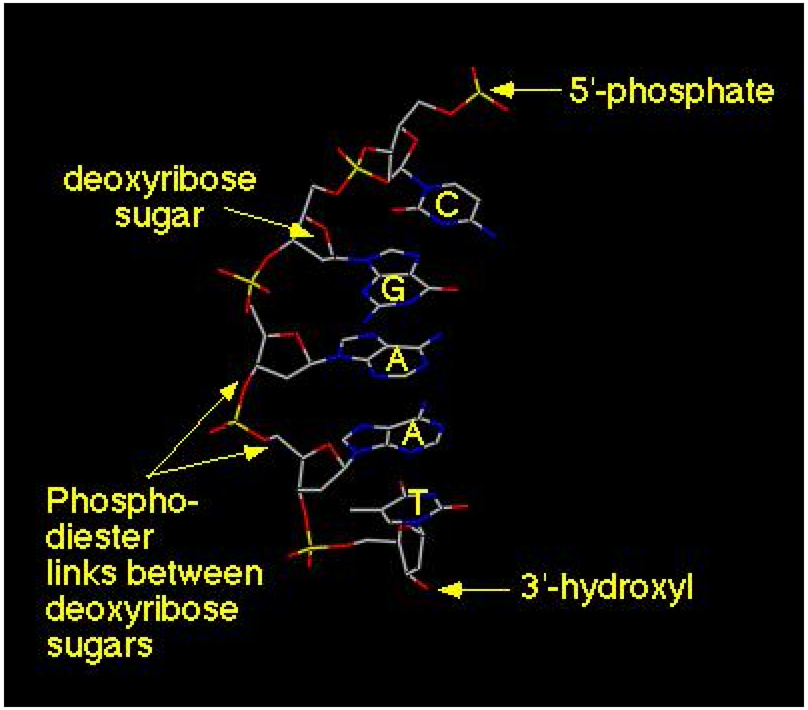
Length of DNA

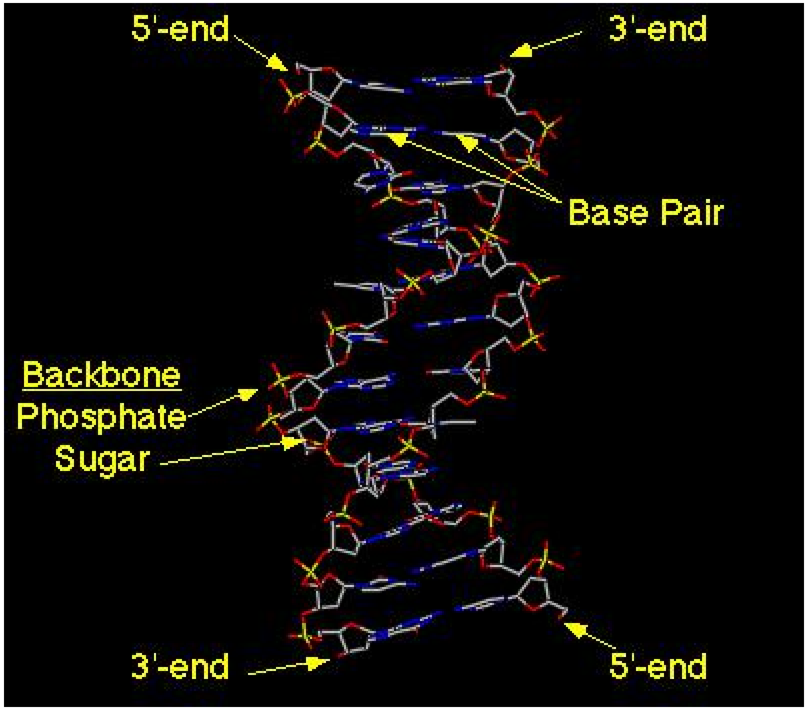


Each DNA polymer = 5 centimeters

DNA (3 billion base pairs) = 2.3 meters long/cell

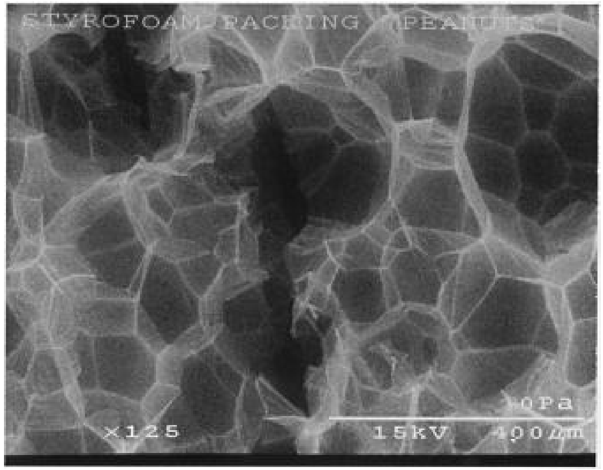
Total length of DNA in a human: 2 x 10^{13} meters





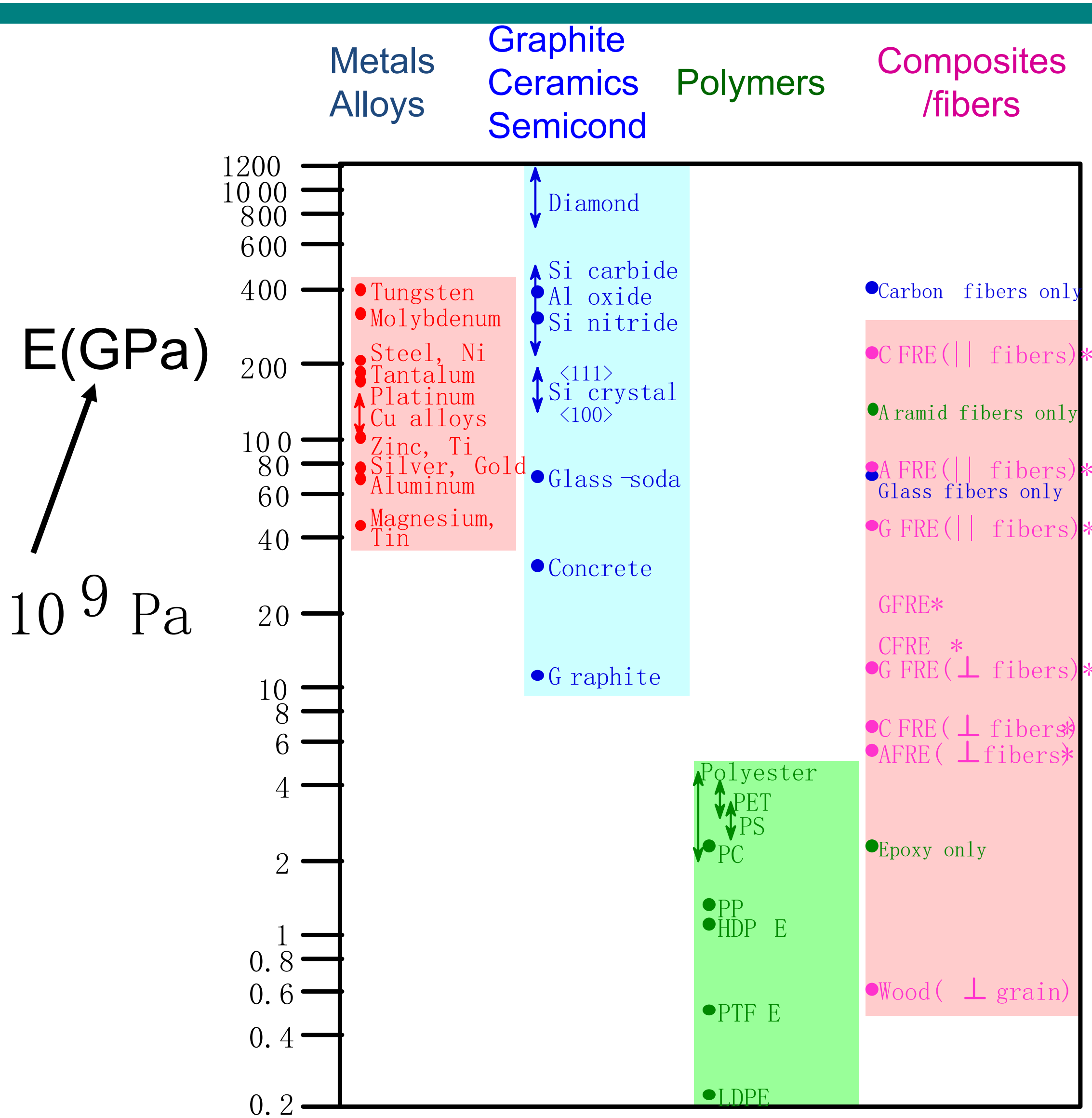
Engineering with Polymers

- Polymers provide a low density structural alternative for some applications
- Are relatively easy to process into numerous forms
- Provide a high volume, often improved replacement for materials derived from living organisms.
- Possess unique properties
- They are often relatively inexpensive.



Styrofoam ®

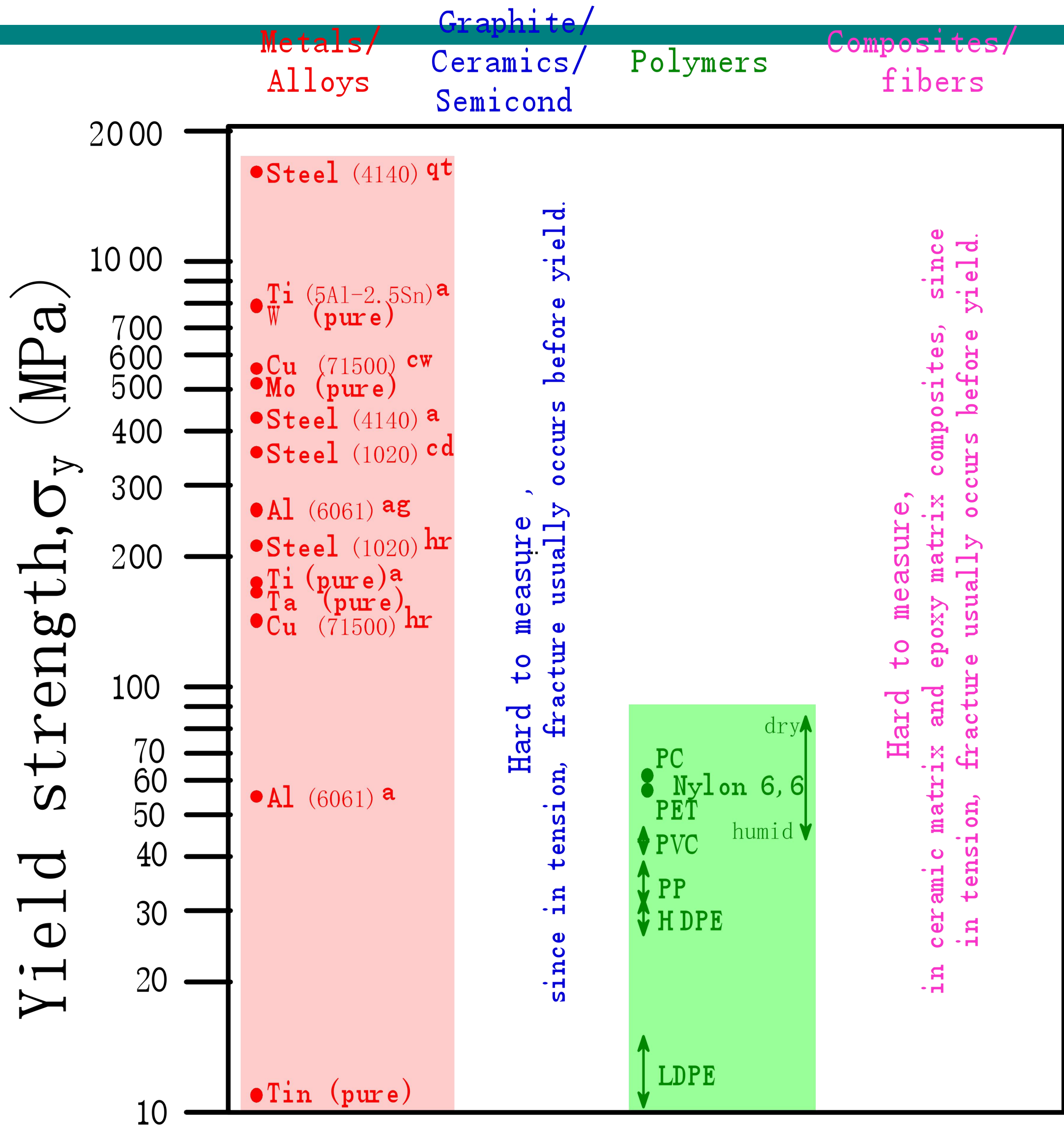
YOUNG'S MODULI: COMPARISON



Based on data in Table B2, *Callister 6e*.

Composite data based on reinforced epoxy with 60 vol% of aligned carbon (CFRE), aramid (AFRE), or glass (GFRE) fibers.

YIELD STRENGTH: COMPARISON



σ_y (ceramics) >> σ_y (metals) >> σ_y (polymers)

Room T values

Based on data in Table B4, *Callister 6e*.

a = annealed
hr = hot rolled
ag = aged
cd = cold drawn
cw = cold worked
qt = quenched & tempered

Why use polymers

- Easy to process
  - Injection molding (thermoplastics)
  - Mold or reaction injection molding (thermosets)
- Cheap
- Lightweight
- Tough
- Flexible
- Transparent (sometimes)
- Insulating (generally)

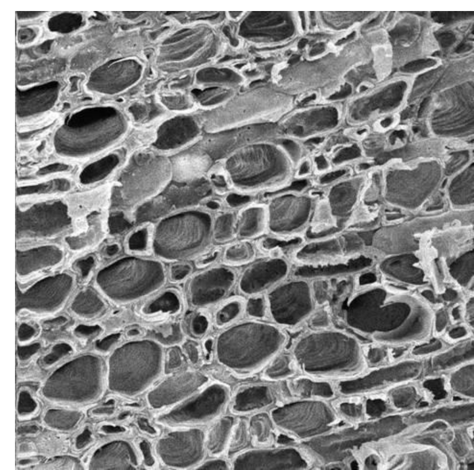
How do we classify polymers?

- By origin
- Physical behavior
- Structure/Architecture
- Application/function
- Polymerization mechanism
- Polymerization chemistry
- Cost

Origin of Polymers

Biopolymers

- Protein: horn, cartilage, hair, hide, ligaments, tusks
- Composite structures: bone, shells
- Plant materials:
  - Cellulose (cotton, sisal, hemp) fiber
  - lignin & cellulose (wood)
  - Chitan (insect & crustacean exoskeletons)



Synthetic Polymers

Coal

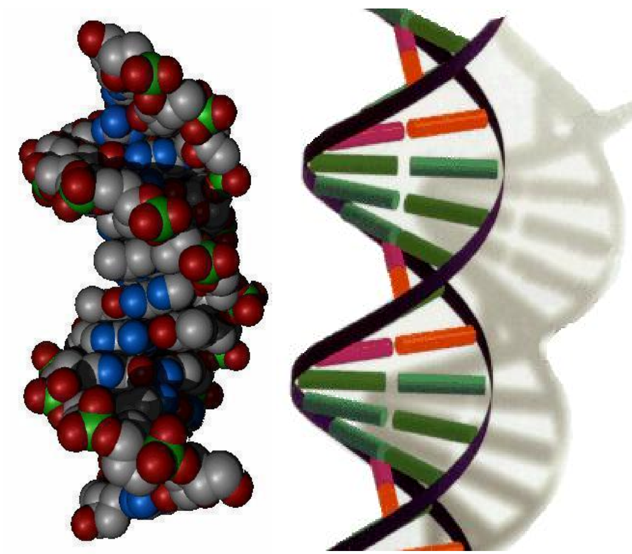
Petroleum

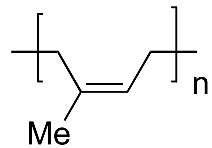
Natural gas

Petroleum from petra oleum (rock oil)

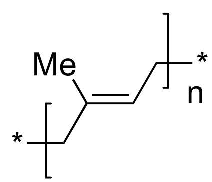
Origins: Two Families of Polymers

Biological Polymers



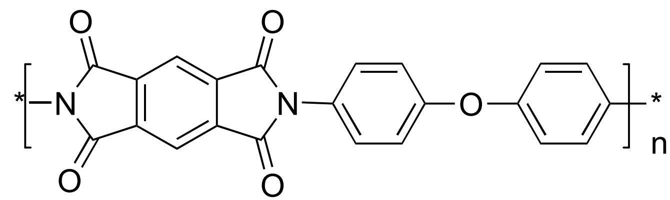


latex rubber

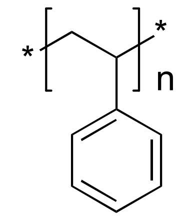


gutta percha

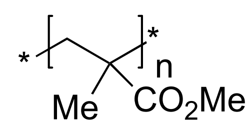
Synthetic



polyimide (PI)



polystyrene



polymethylmethacrylate (PMMA)

Physical Behavior & Architecture

- Thermoplastics

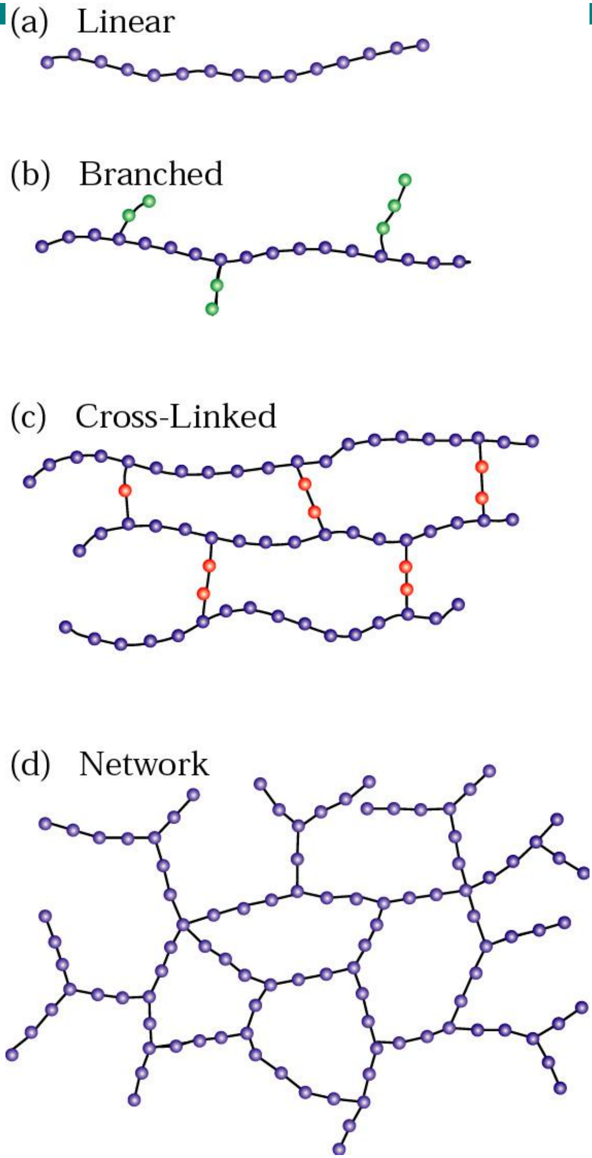
  Polystyrene
  Polyvinylchloride

- Elastomers

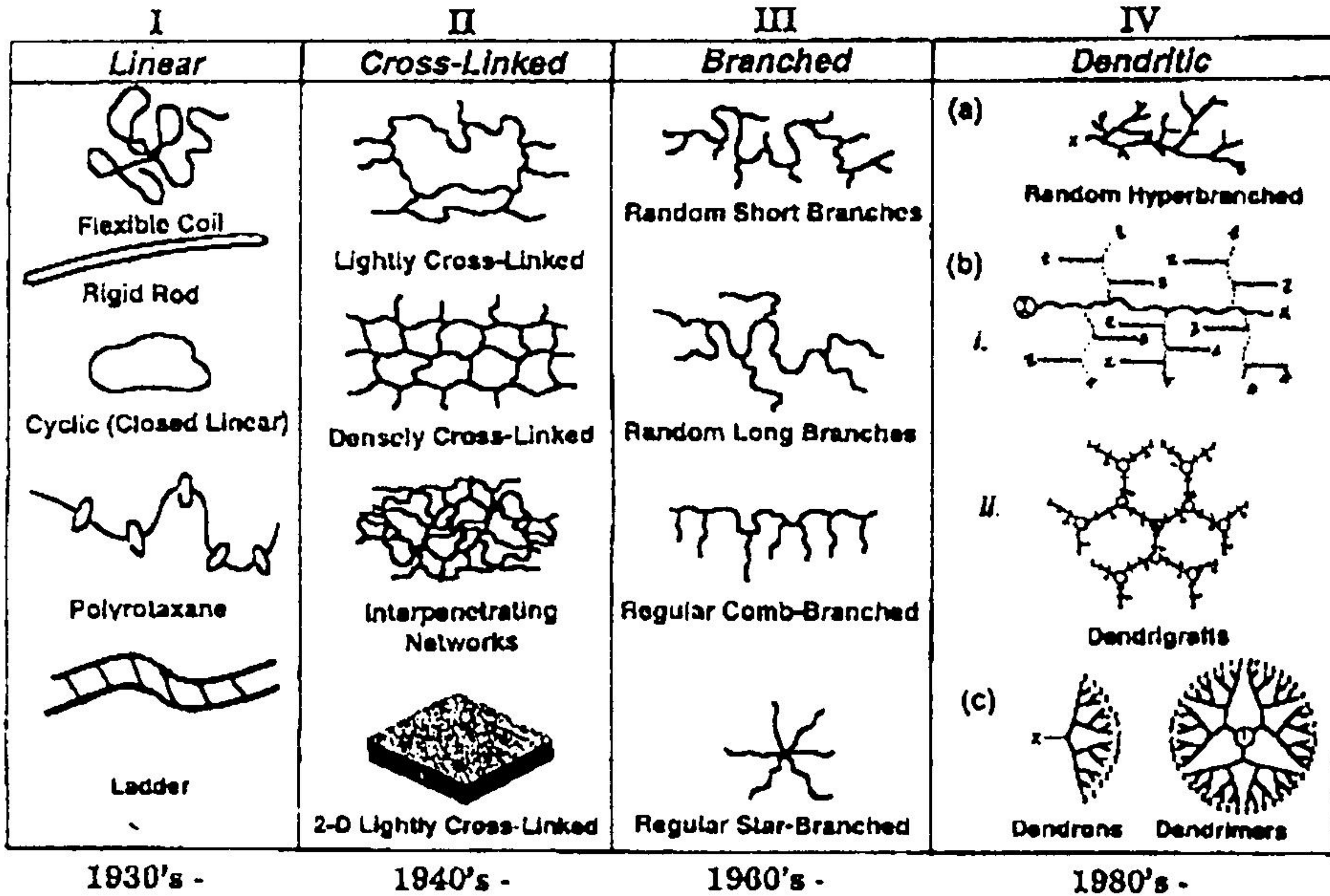
  Synthetic rubbers
  Poly-cis-isoprene

- Thermosets

  Phenolic Resins
  Melamines
  epoxies



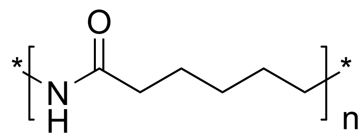
Major Macromolecular Architectures



Source: R. Esfand, D.A. Tomalia, A.E. Beezer, J.C. Mitchell, M. Hardy, C. Orford, Polymer Preprints, **41** (2), 1324 (2000)

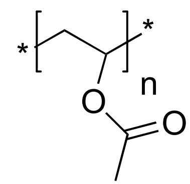
Applications/Function

- Structural



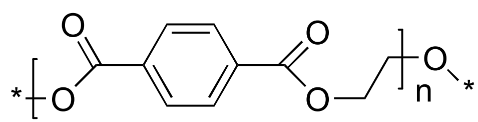
Nylon-6

- Coatings



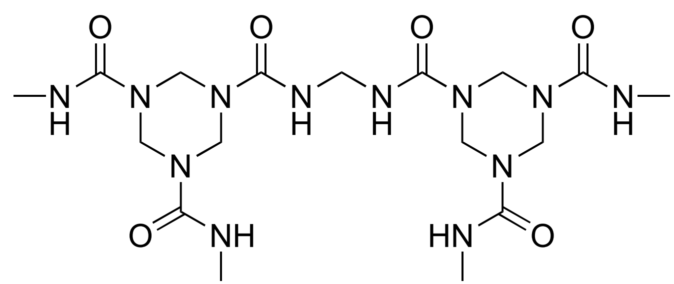
Poly(vinyl acetate) or PVA

- Fibers



Poly(ethylene terephthalate) or PETE

- Adhesives



Urea-Formaldehyde

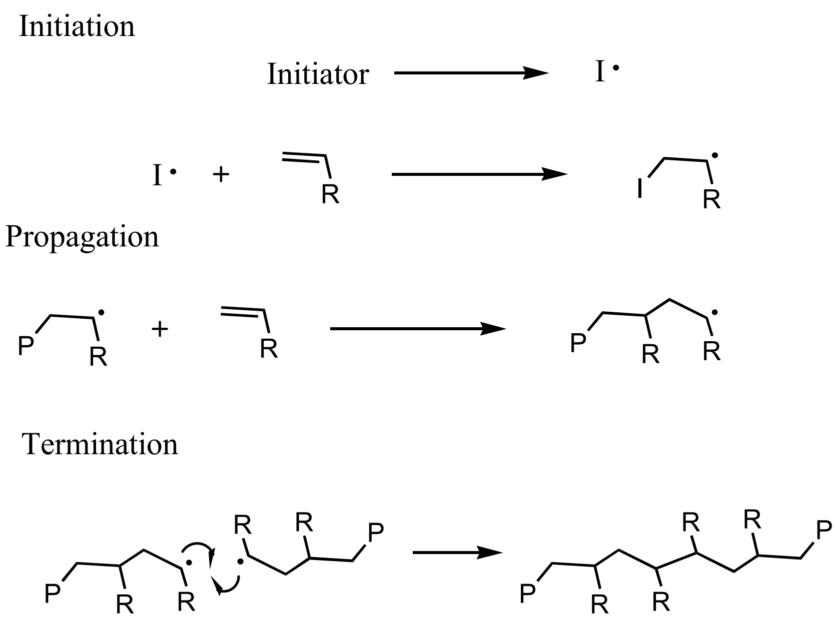
Taxonomy by polymerization mechanism

Chain Growth Mechanism

- Free radical
- Anionic
- Cationic
- Ring opening

Step growth

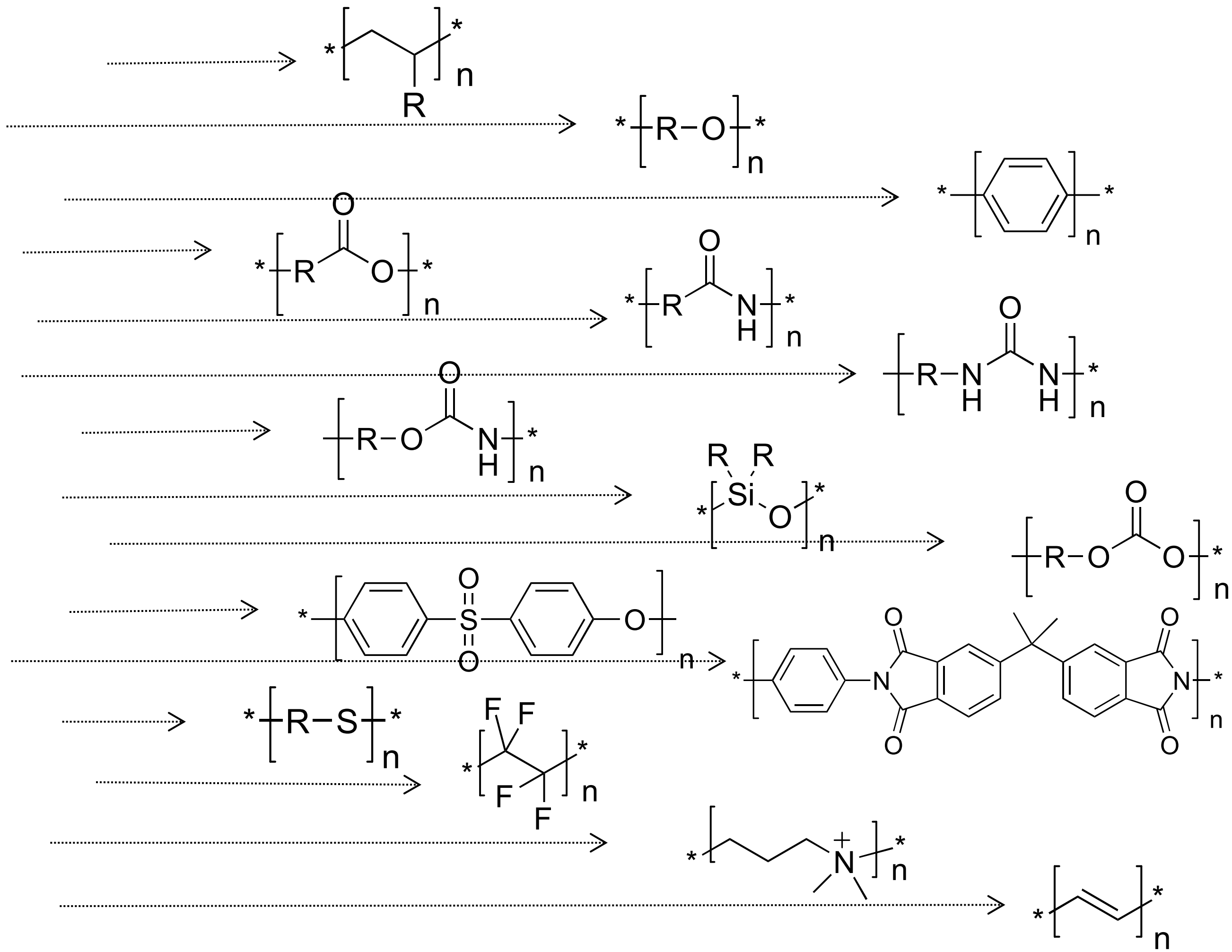
- Condensation
- Metathesis



Free radical chain mechanism

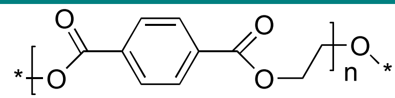
Polymer Functionality

Vinyl Polymers
Polyethers
Polyarylenes
Polyesters
Polyamides
Polyureas
Polyurethanes
Polysiloxanes
Polycarbonates
Polysulfones
Polyimides
Polysulfides
Fluoropolymers
Polyionomers
Polyacetylenes



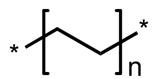
Recycling symbols

1 PETE

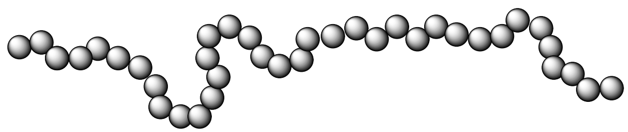


Poly(ethylene terephthalate) or PETE

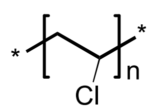
2 HDPE



high density polyethylene

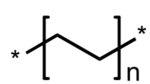


3 V

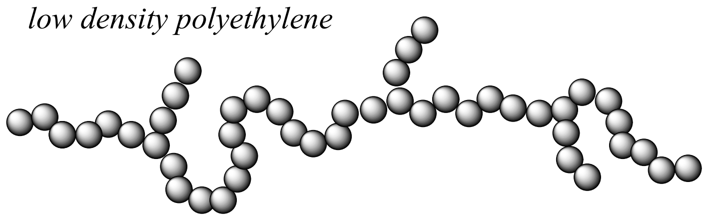


polyvinyl chloride

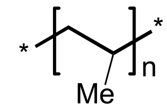
4 LDPE



low density polyethylene

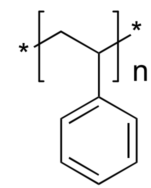


5 PP



poly(propylene)

6 PS



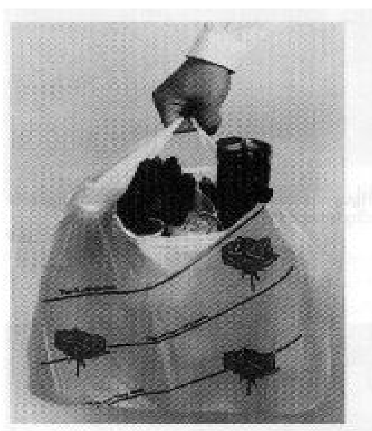
polystyrene

7 Other

Not recyclable

Cost: Commodity (Amorphous) Thermoplastics

- Four high volume thermoplastics and applications:
  - Polyethylene (PE): Grocery bag, 55-gallon drum, lawn furniture
  - Polypropylene (PP): Washing machine agitator, carpet
  - Polyvinylchloride (PVC): Irrigation pipe, wire insulation
  - Polystyrene (PS): Toys, pipes, packing material (Styrofoam)

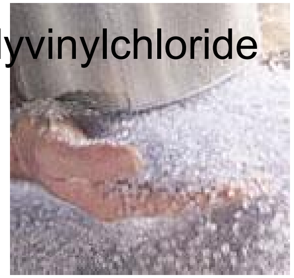


Polyethylene

Polypropylene

Polystyrene

Polyvinylchloride



- Low cost, temp. resistance and strength • Good dimensional stability
- Bonds well • Typically, but not always, transparent

Some History: First there were Bio-Polymers

Animal Hides (Proteins): Fiber & Films
Ligaments (Collagen): Hinges
Silk Fibers (Protein): Fibers
Plant Fibers (Cellulose): Fibers

Bison-Hide teepee

Yucca-fiber sandals

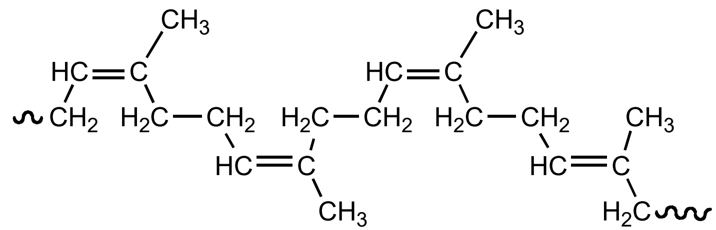
Ivory lunar cycle charts

<u>Structural Materials: High Modulus & Strong</u>
Wood (Cellulose & Lignin): S
Antlers (Keratin): Tools, jewelry & weapons
Horn (Keratin): Tools, jewelry & weapons
Tusks (enamel & dentin): Tools, jewelry & weapons

Key Figures in Polymer History:

Charles Goodyear
(1800 - 1860)

Invented vulcanization
of rubber in 1839



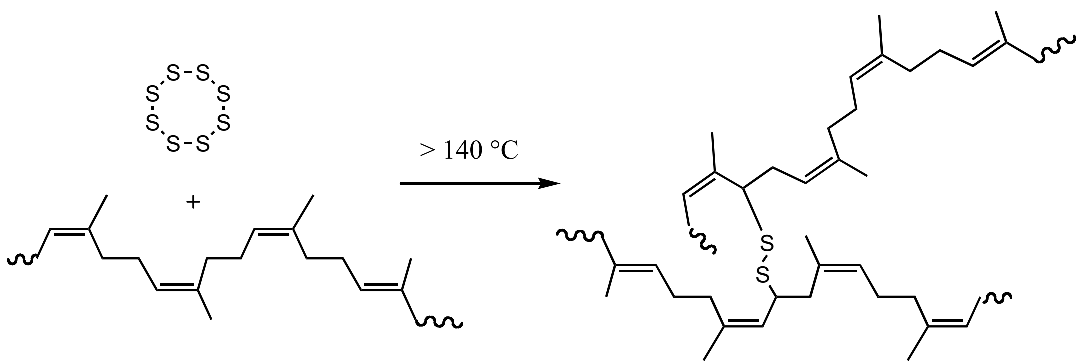
Poly-*cis*-isoprene

IUPAC: *cis*-poly(1-methyl-1-butene-1,4-diyl)

Elastomer:

50% of Rubber tires

Latex rubber gloves

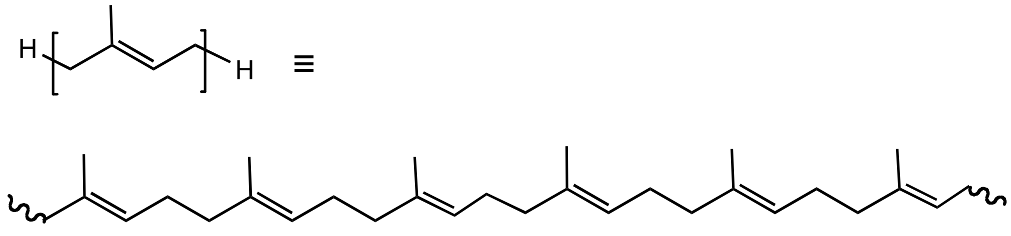


Enabled commercialization of natural rubber

Gutta Percha

William Montgomerie (1840’s)

Saw usefulness of gutta percha



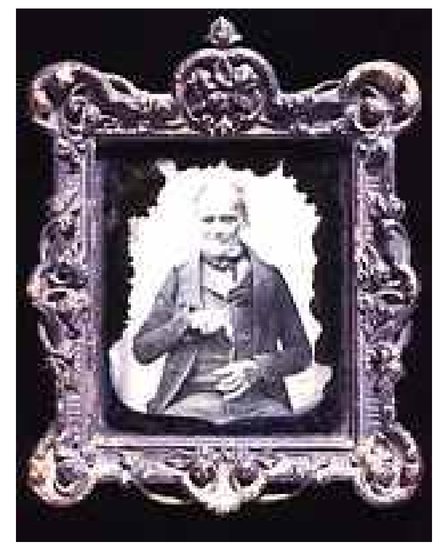
trans-Polyisoprene or Gutta percha

IUPAC: *trans*-poly(1-methyl-1-butene-1,4-diyl)

Thermoplastic:

Golf ball covers

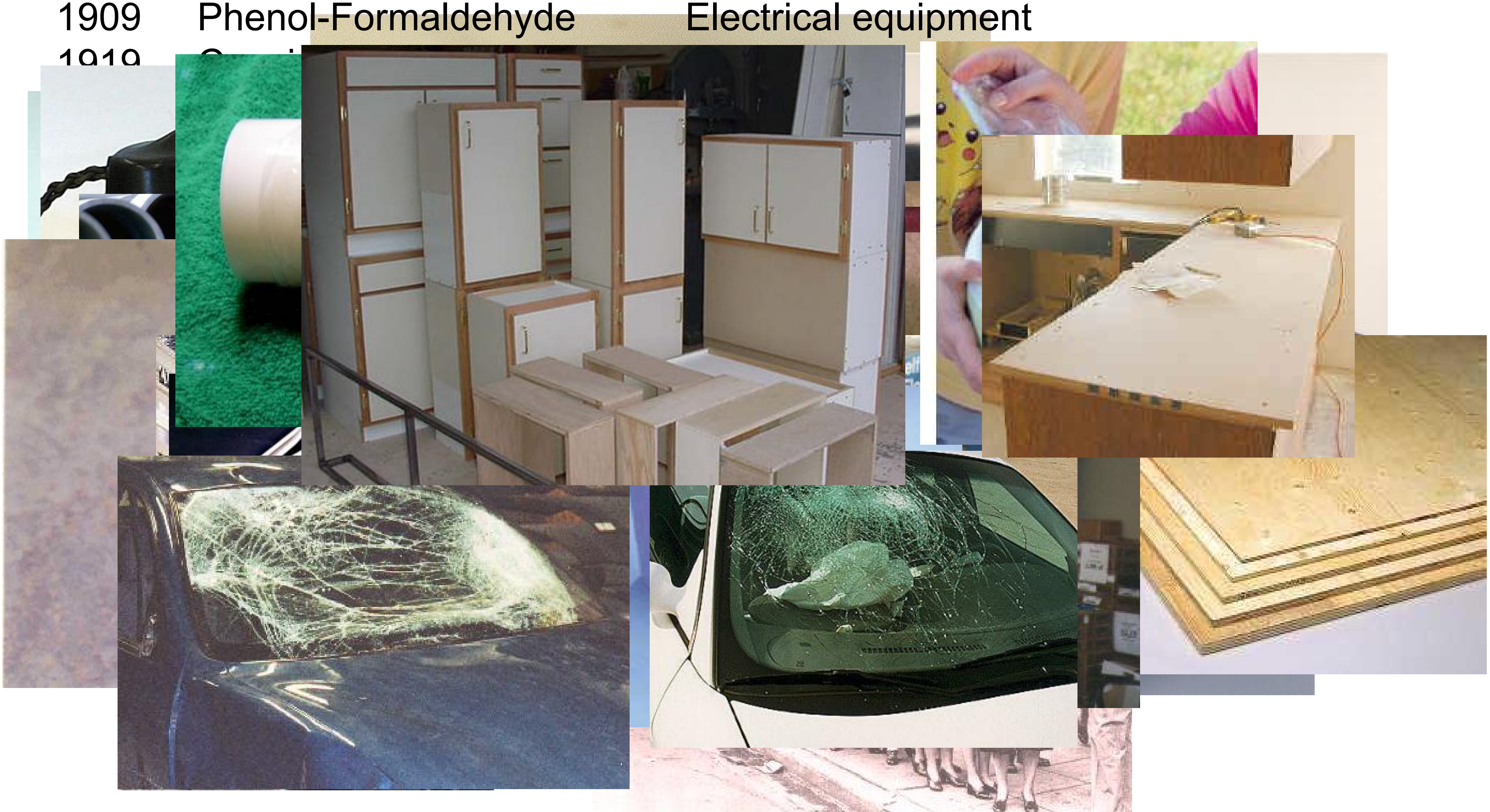
Wire coating (until 1940’s)



Gutta percha (GP), also known as balata, is a natural thermoplastic and is of fundamental importance in the history of the plastics industry.

History of Polymers

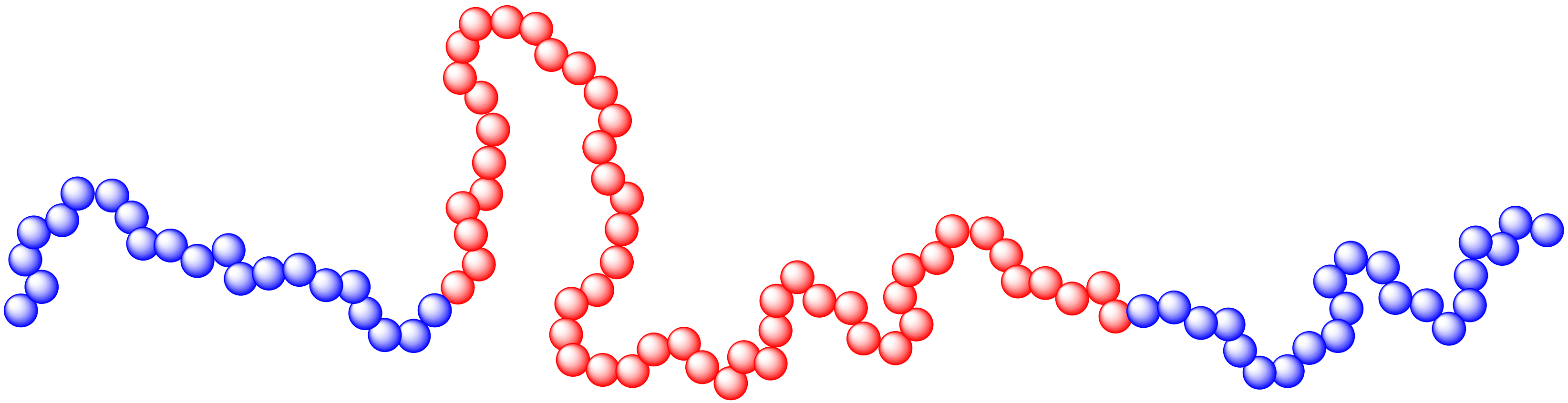
Date	Material	Example Use
1868	Cellulose Nitrate	Figurines
1909	Phenol-Formaldehyde	Electrical equipment

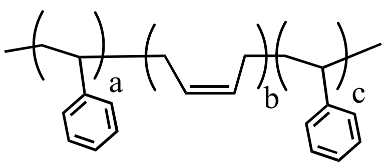


History of Polymers

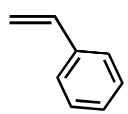
Date	Material	Example Use
1942	Polyester	Clothing, Boat hulls
1942	Polyeth	ugs
1943		ts, Non-stick liners
1943		ils
1947		
1948		
1954		heels
1956		
1957		
1957		

Block Copolymers





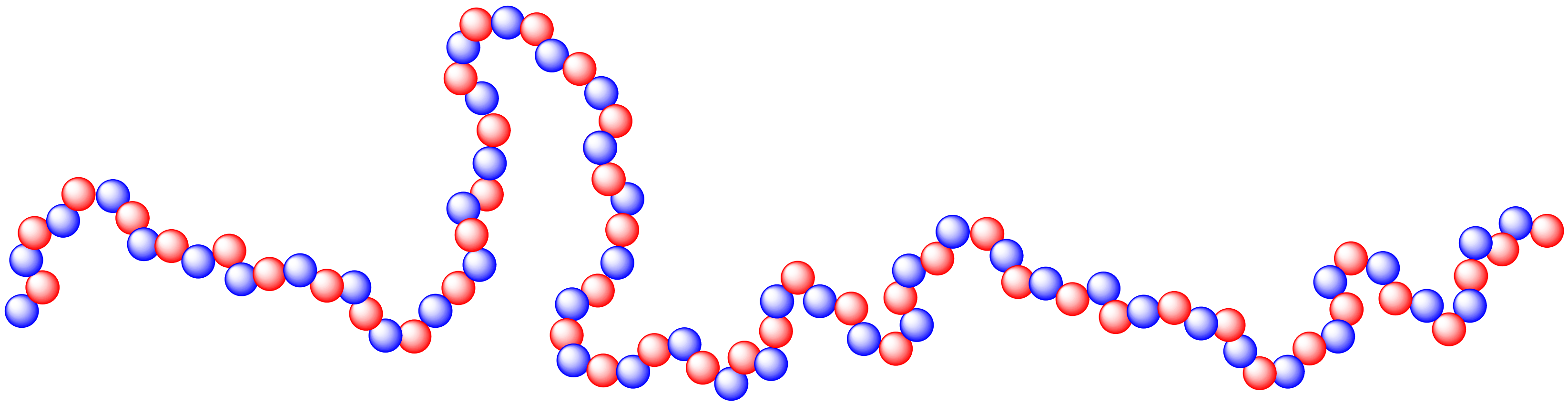
Polystyrene-block-poly-1,4-butadiene-block-polystyrene

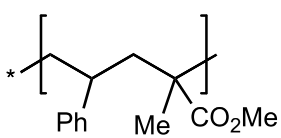


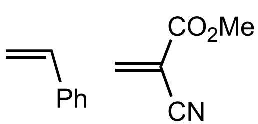
Block-copolymer[styrene-butadiene-styrene]

SBS

Alternating Copolymers



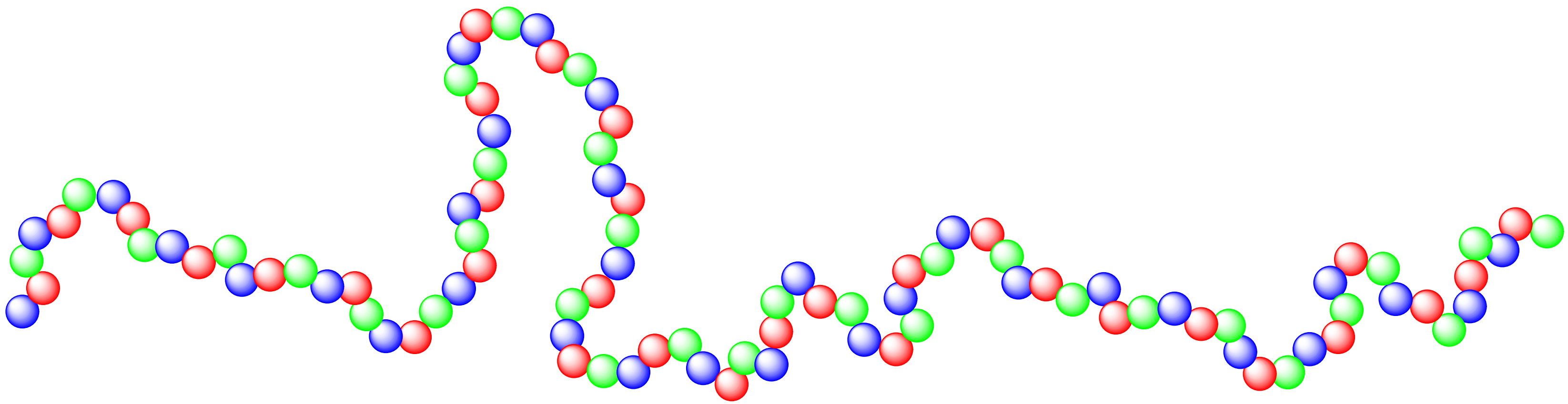


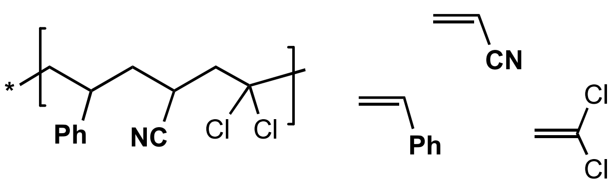


Poly[styrene-alt-(methyl methacrylate)]

Alt-copoly[styrene/methyl methacrylate]

Alternating Copolymers

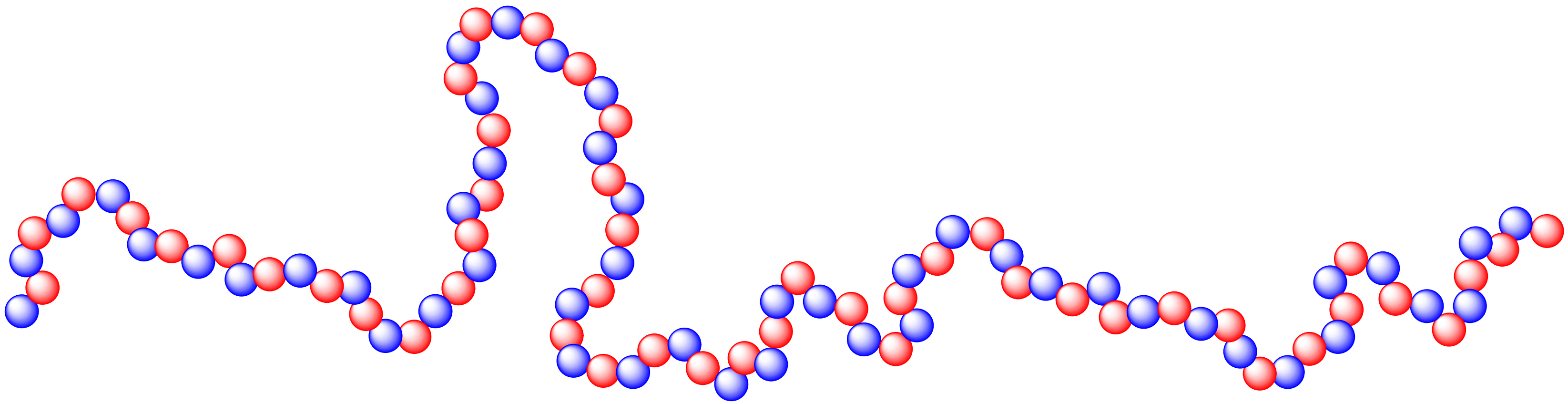




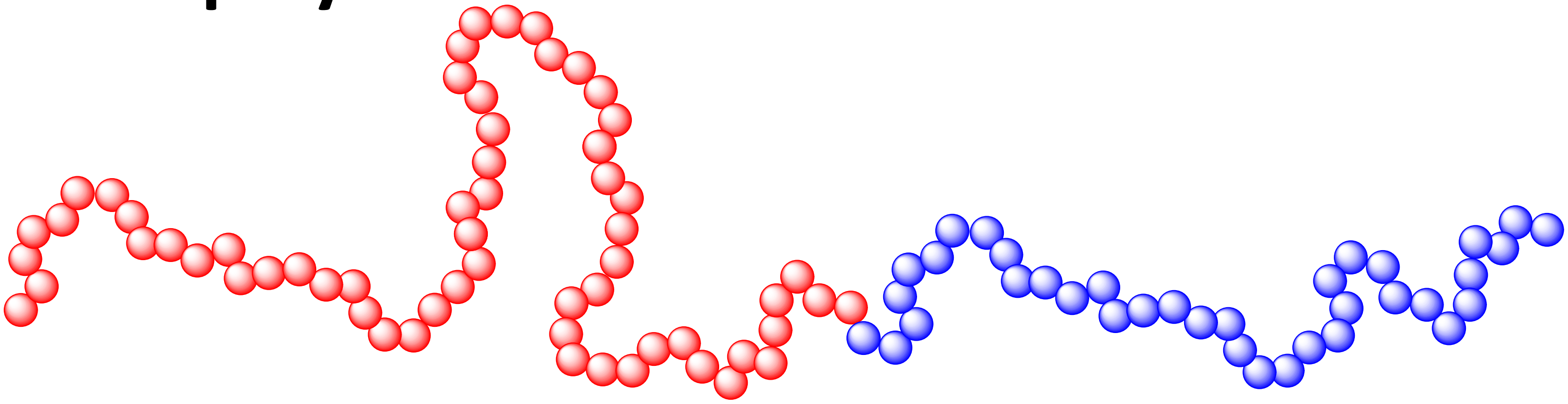
Poly[styrene-alt-(acrylonitrile)-alt-(vinylidene dichloride)]

Alt-copoly[styrene/acrylonitrile/vinylidene dichloride]

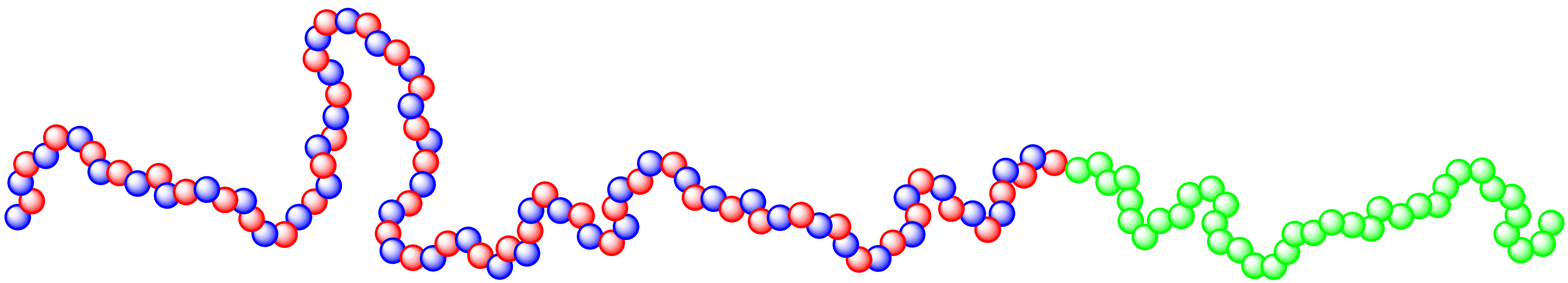
Alternating Copolymers

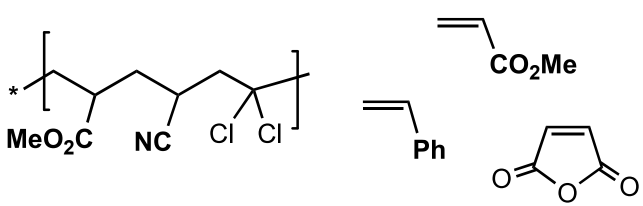


Block Copolymers



Block & Alternating Copolymer

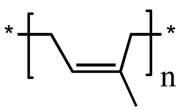
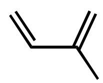
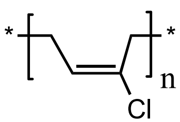
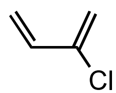
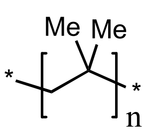
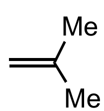




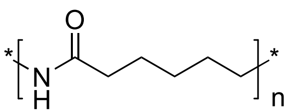
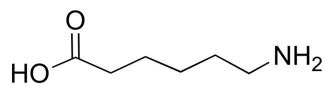
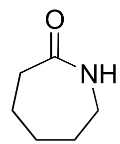
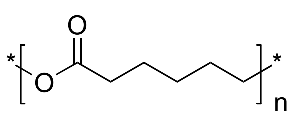
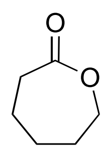
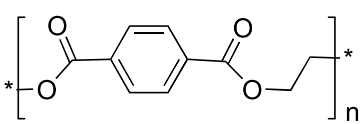
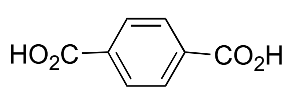
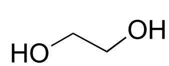
Poly[methyl acrylate-block-(poly(maleic anhydride)-alt-styrene)]

Block-copoly[alt-co(styrene/maleic anhydride)methyl acrylate]

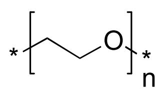
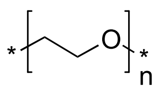
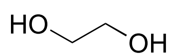
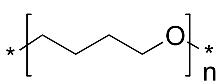
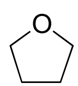
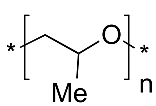
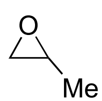
Nomenclature of Elastomers

Structure	Monomer	Common Name	IUPAC	Trade name
		*cis*-Polyisoprene	*cis*-poly(1-methyl-1-butene-1,4-diyl)	latex
		Polychloroprene	poly(1-chloro-1-butene-1,4-diyl)	Neoprene
		Polyisobutylene	poly(1,1-dimethyl-ethene-1,2-diyl)	Butyl Rubber

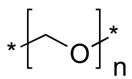
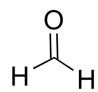
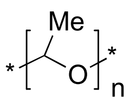
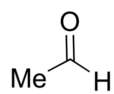
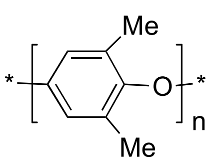
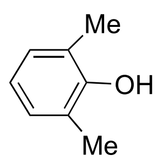
Nomenclature of Thermoplastics

Structure	Monomer	Common Name	IUPAC	Trade name
	 6-aminohexanoic acid  azepan-2-one or caprolactam	poly(6-hexanomide) polycaprolactam	poly(imino(1-oxohexamethylene))	Nylon-6
		polycaprolactone	poly(oxy(1-oxohexamethylene))	
	 	poly(ethylene terephthalate)	Poly(oxyethylene -oxyterephthaloyl)	PETE

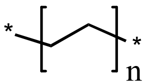
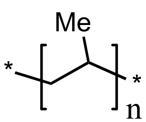
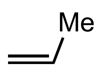
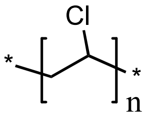
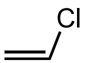
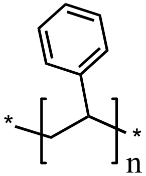
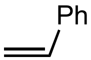
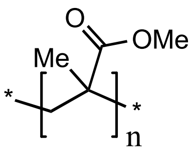
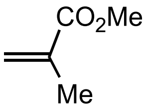
Nomenclature of Polyether Thermoplastics

Structure	Monomer	Common Name	IUPAC	
	 ethylene oxide or oxirane	polyethyleneoxide	polyoxyethylene	
	 ethylene glycol	poly(ethylene glycol)	polyoxyethylene	**PEG**
	 tetrahydrofuran	poly(tetrahydrofuran)	poly(oxytetramethylene)	
	 propylene oxide	poly(propylene-oxide)	poly(oxy(1-methylethylene)	

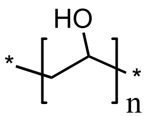
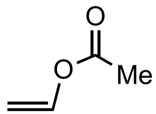
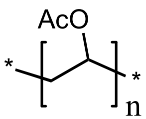
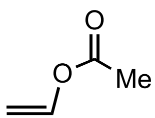
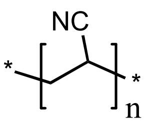
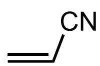
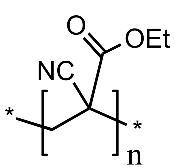
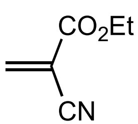
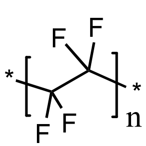
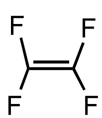
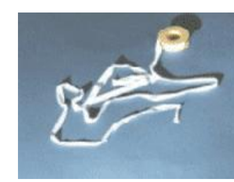
Nomenclature of Polyether Thermoplastics

Structure	Monomer	Common Name	IUPAC	
	 formaldehyde	poly(formaldehyde)	poly(oxymethylene)	
	 acetaldehyde	poly(acetaldehyde)	poly(oxyethylidene)	Delrin
		poly(phenyleneoxide)	poly(oxy-2,6-dimethyl-1,4-phenylene)	

Nomenclature of Vinyl Thermoplastics

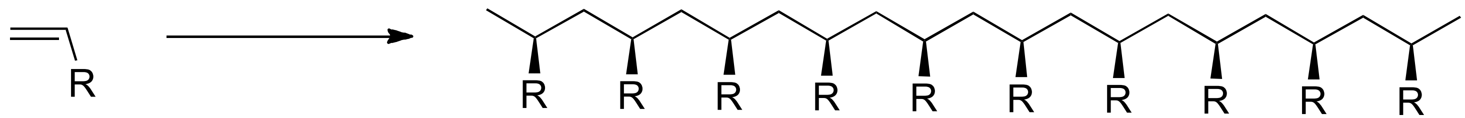
Structure	Monomer	Common Name	IUPAC	Trade Name
		Polyethylene	poly(ethylene)	PE
		Polypropylene	poly(propylene)	PP 
		Polyvinyl chloride	poly(1-chloroethylene)	PVC
		Polystyrene 	poly(1-phenylethylene)	PS
		Polymethyl methacrylate	poly(1-(methoxycarbonyl) -1-methylethylene)	PMMA

Nomenclature of Vinyl Thermoplastics

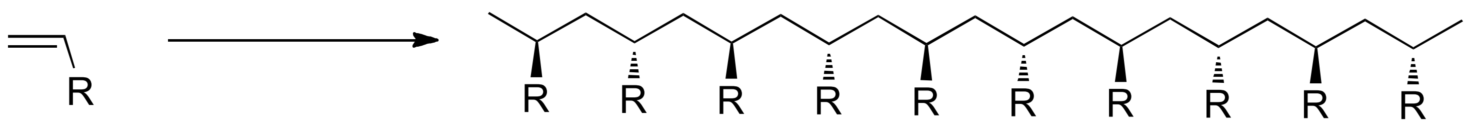
Structure	Monomer	Common Name	IUPAC	Trade Name
		Polyvinyl alcohol	poly(1-hydroxyethylene)	PVA
		Polyvinyl acetate	poly(1-acetoxyethylene)	
		Polyacrylonitrile	poly(1-cyanoethylene)	PAN 
		Poly(ethyl cyanoacrylate)	poly(1-cyano-1-(ethoxy carbonyl)ethylene)	Super glue
		Poly(tetrafluoroethylene)	Poly(tetrafluoroethylene)	 Teflon

Tacticity: How groups are arranged along polymer

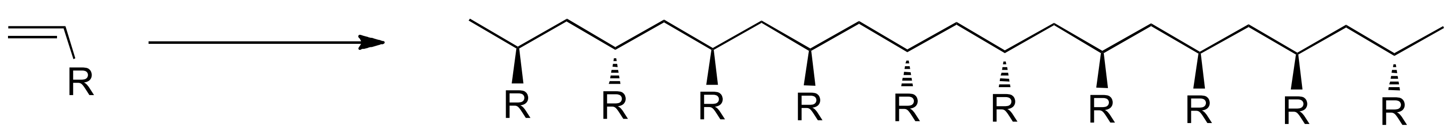
Vinyl Monomers



isotactic



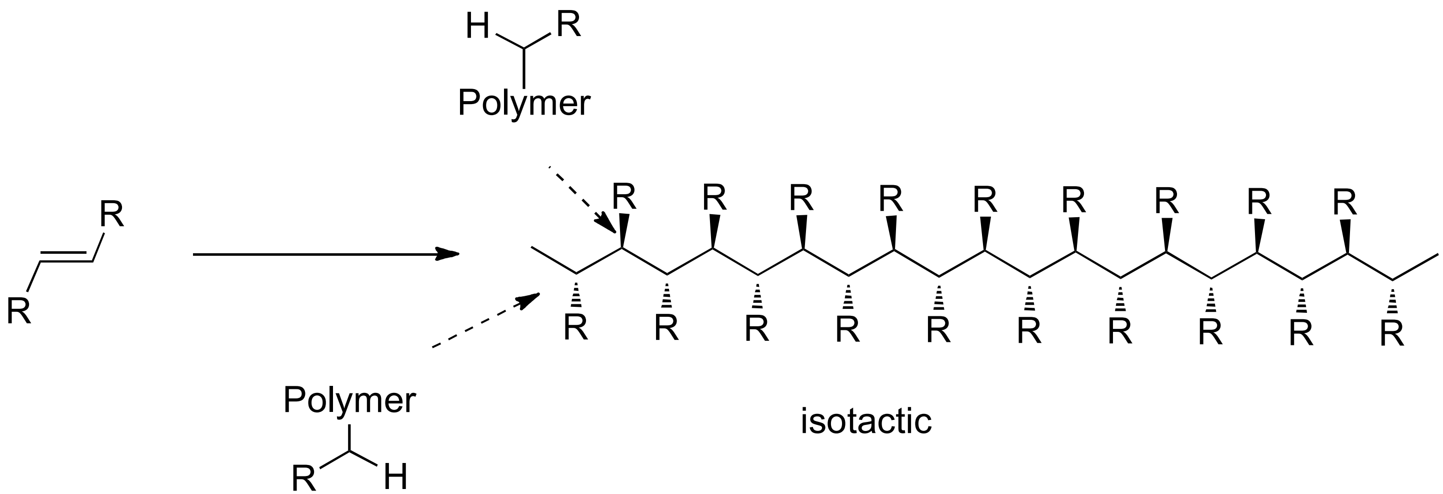
syndiotactic



atactic

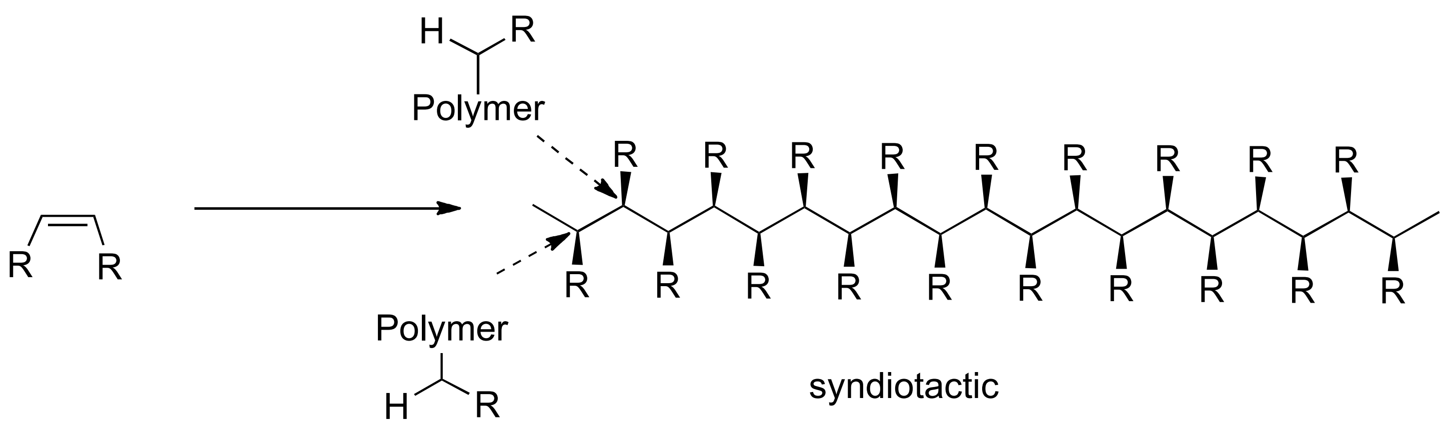
Isotactic and syndiotactic pack into lattices easier: crystalline

Tacticity: disubstituted monomers



isotactic

Looksing at indicated atoms from the left



syndiotactic

HOMEWORK

"Polymers in our daily lives"

Write an assay